Iter1.2 Research

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User Research

We wanted to understand how students struggle with procrastination in daily life, not only in theory. We ran a google form survey with around 23 respondents from our target segment, polytechnic and university students in Singapore aged 17–25. We distributed the survey through Telegram group chats, Instagram stories, focusing on NUS, NTU, SMU, and local polytechnics.

From the survey and earlier proposal work, over 80% of respondents reported they struggle to stay consistent with habits such as studying, exercising, or reading. This matches evidence that procrastination is common among university students in Singapore and is linked to stress and poorer outcomes (Yeo and Chiam 2025). Many students said external pressure from deadlines, grades, or expectations from others drives them more than private reminders. This aligns with behavioral research on precommitment and loss aversion, where people are more responsive to predefined consequences and the fear of losing something than to abstract future gains (The Decision Lab 2021).

Social accountability can be a supporting factor. Roughly three quarters of students (from survey) agreed that having friends hold them accountable would improve their consistency, and a similar proportion said they felt comfortable sharing habit progress with friends in a controlled environment. This supports FoE's core idea of involving friends directly in the accountability loop instead of keeping productivity private.

Students also struggle with the long-term academic rewards. Long term goals such as exams that are months away feel vague, while apps like TikTok or games provide instant satisfaction. Behavioral work on temporal discounting (cognitive bias where people value rewards less the further in the future they are) shows that people heavily discount future rewards relative to immediate ones, which fits what students described (The Decision Lab 2021). One cliche statement was "I know it is important, but it does not feel urgent until the night before." FoE's near term, pre-agreed consequences are meant to create pressure in this gap.

When we asked about consequences, most respondents preferred light, social, and effort based penalties over large financial ones. They were open to "social blunders," small physical tasks such as push ups, or status changes in the app, but were reluctant to risk losing significant real money.

Impact on problem and solution

- The main problem is not a lack of productivity apps, but a lack of enforced accountability and immediate stakes that matter emotionally to students.

- Results support placing social accountability and friend verification at the centre of FoE, instead of building another solo to do list or timer.

- Findings support a library of predefined, safe, and playful consequences, including social, physical, and status based options, rather than heavy reliance on real money penalties.

- A social first, gamified interface with a feed, profiles, and streaks fits how students already use mobile apps and social platforms.

Appendix

1. How often do you struggle to stay consistent with habits such as studying, exercising, or reading?
2. What currently motivates you the most to stay consistent? (Select all that apply)
3. Would having friends hold you accountable increase your consistency with habits
4. How comfortable would you be with sharing habit progress with friends?
5. Do you find gamified systems (points, streaks, challenges, leaderboards) motivating?
6. Would real consequences (e.g. fun social penalties, small donations, light physical tasks) motivate you to complete tasks?
7. Which types of consequences would you find acceptable? (Select all that apply)
8. How likely would you be to use this type of app (a gamified accountability app with social enforcement and consequences)?
9. What would make you more likely to use this type of app?

Below is the summary of the responses for each question:

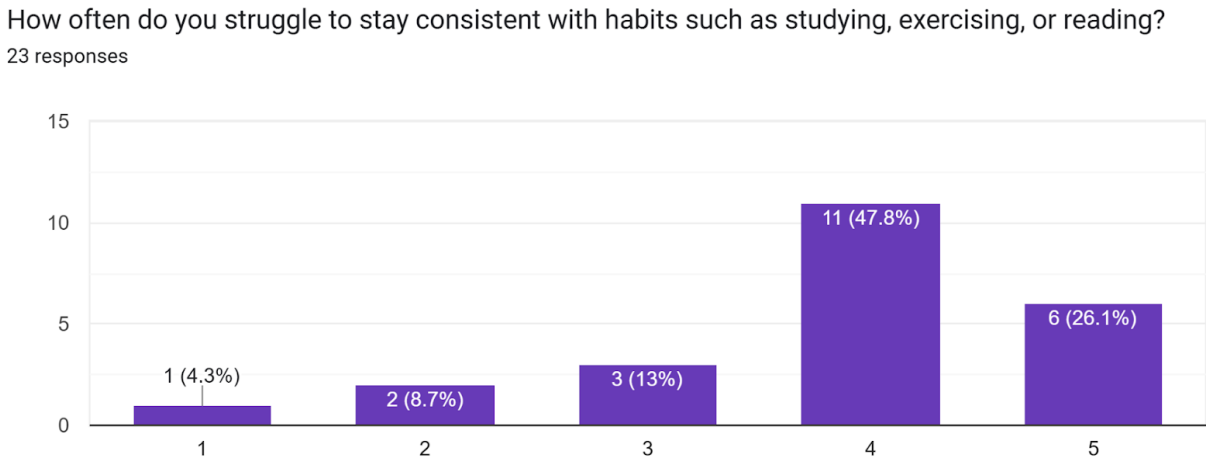

Figure 5. Survey result on staying consistent with habits

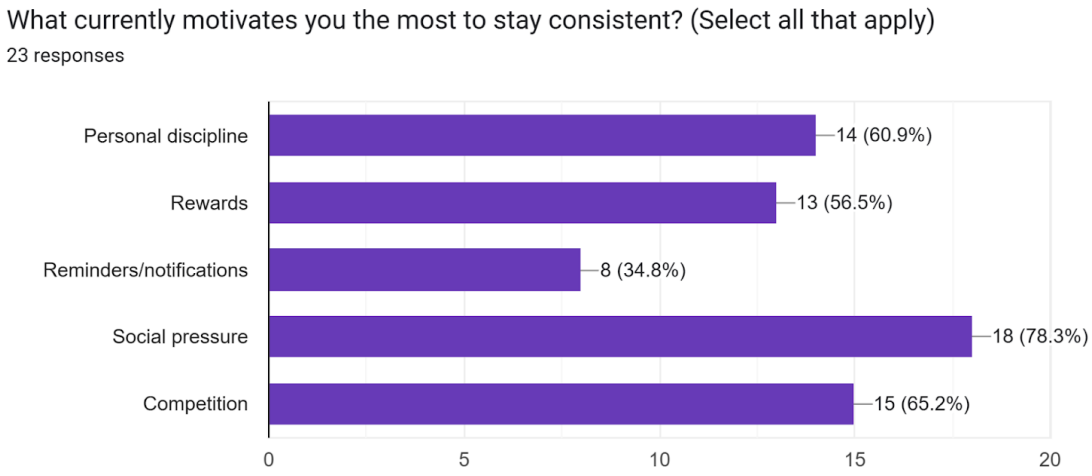

Figure 6. Survey results on what motivations push people to stay consistent

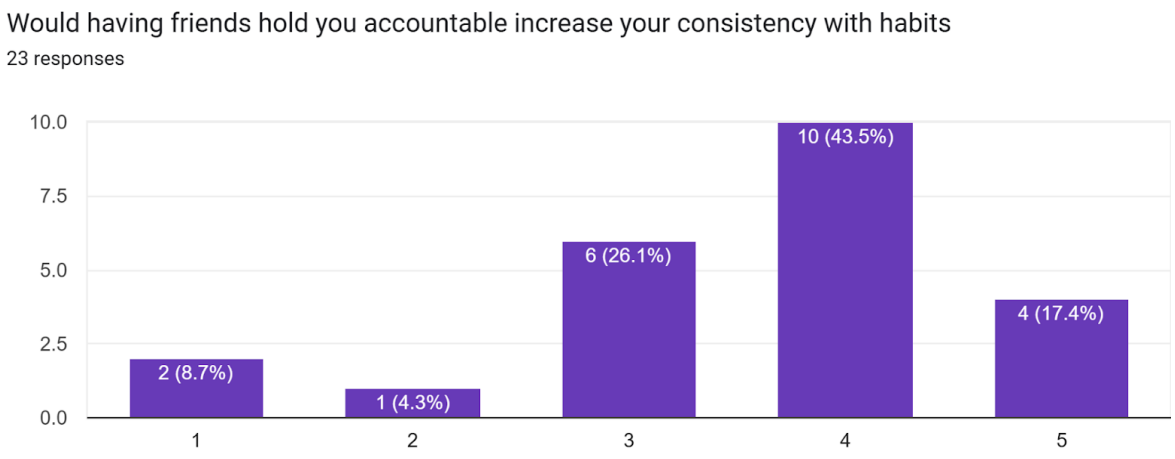

Figure 7. Survey result on the impact of friends in your accountability

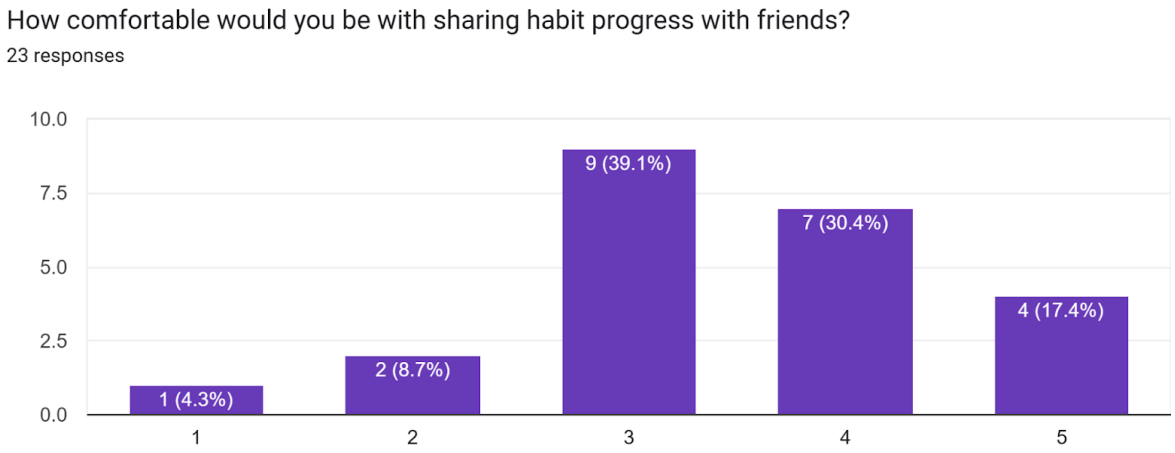

Figure 8. Survey results on how comfortable are people to share habit progress with friends

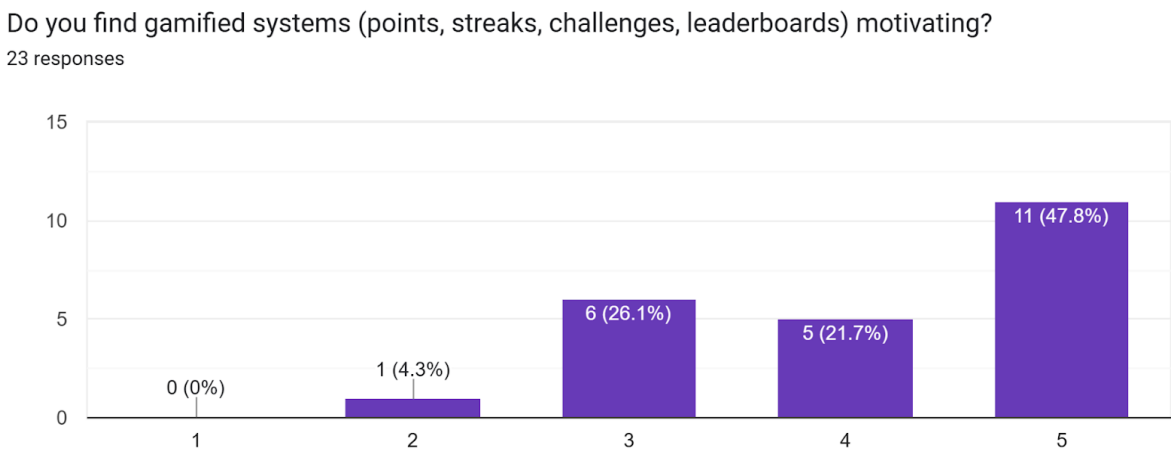

Figure 9. Survey result on whether they find gaming systems motivating

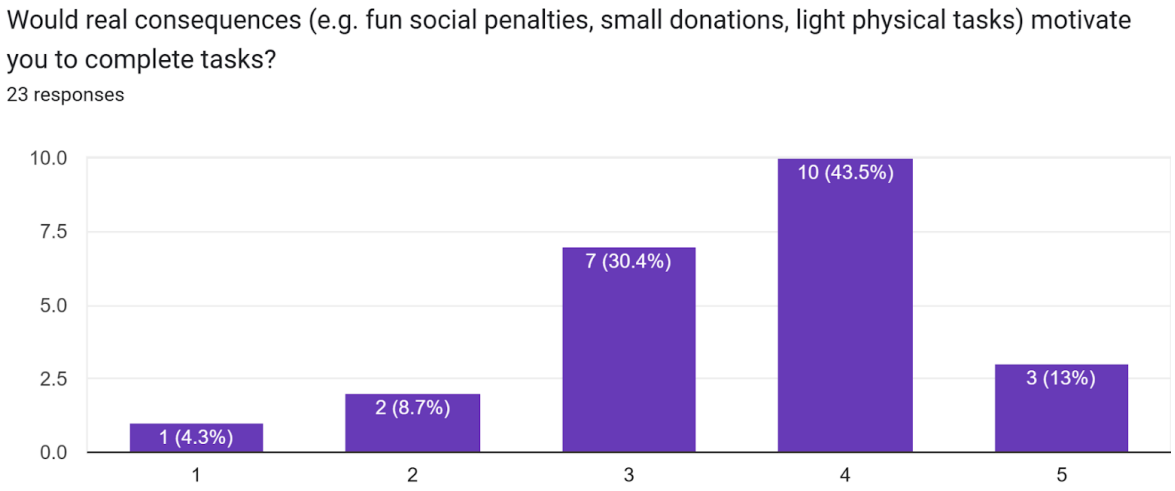

Figure 10. Survey result on whether or not real-consequences motivates you to complete task

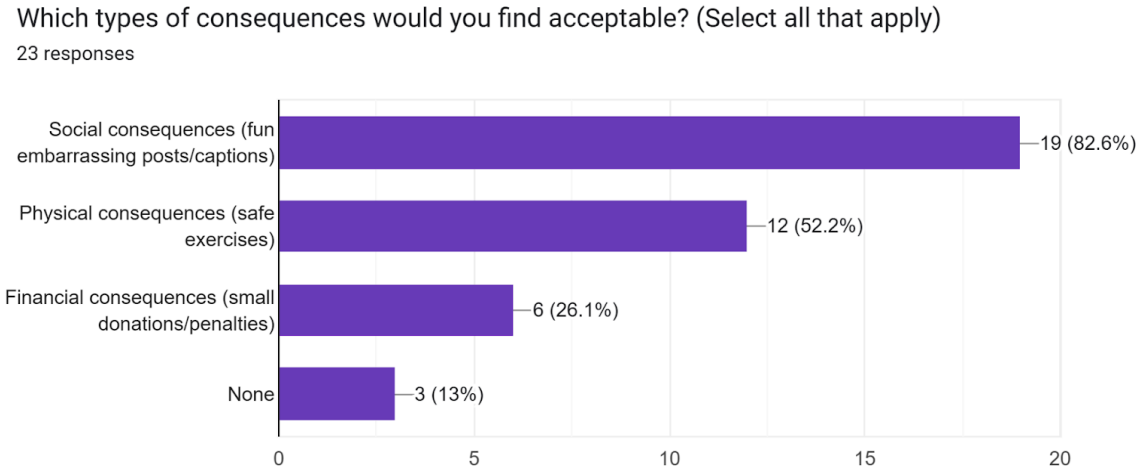

Figure 11. Survey result on which type of consequence is more acceptable

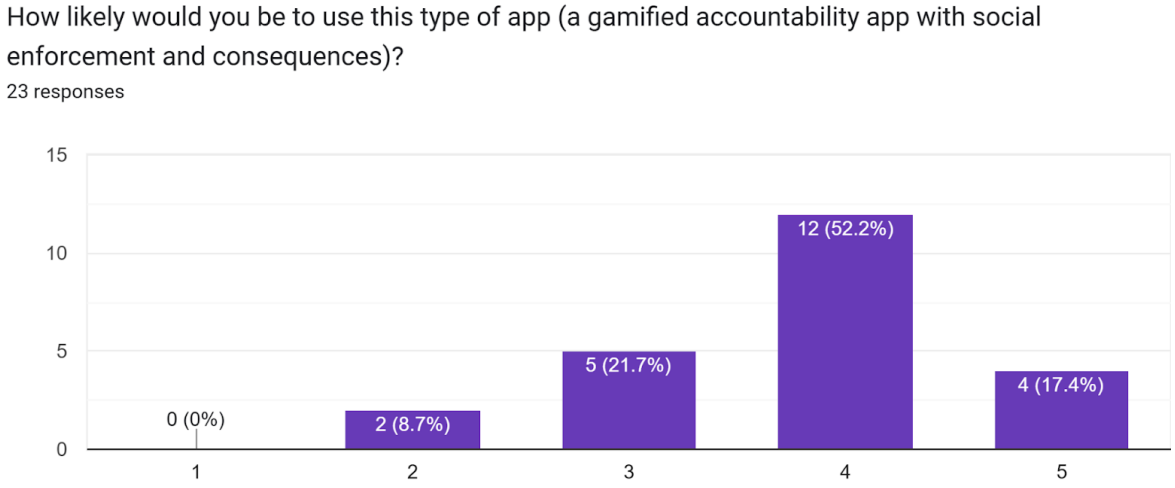

Figure 12. Survey result on whether they would use this kind of app

For the last question which was open-ended, there were a few common themes from the responses, mainly suggesting the idea that they were more likely to use the app if there were rewards also included alongside the punishments, if their friends were also using the app, and if it was easy to use/can be integrated into their daily routine. From all the responses for the survey, there is a clear desirability for the app we are proposing. Firstly, most respondents have struggled with staying consistent and being motivated, and what currently motivates them the most is social pressure. In regards to the social aspect of the app, most of the respondents agreed that having friends hold them accountable would improve their habits and consistency, and were for the most part also okay with sharing their habit progress with their friends. Another

point is that they also think that some consequences could help motivate them to complete their tasks, with the most preferred punishment being social consequences that are more lighthearted. Finally, there is a positive reception to using a gamified system to help motivate them to use the app, and overall there is decent interest in using our proposed app if it were to be implemented.

Technology Research

FoE is designed for the environments students use most often, their phones, in the same attention space as Telegram and Instagram. A social accountability tool that relies on pact creation, timely notifications, and fast photo or video uploads works best as a mobile first product.

Platform and stack

So far we reviewed 3 different approaches, native iOS and Android development, a web only product, and cross platform frameworks such as Flutter or React Native. Given the team's prior full stack experience and the need to support both platforms quickly, a cross platform mobile framework such as Flutter or React Native is a practical fit. This supports a single codebase while still offering a familiar, responsive app experience close to other social and productivity apps used by students (Vorecol 2024).

Backend and data

For working on the MVP, running and maintaining a full custom server stack from day one would be heavy. A serverless style backend with managed authentication, database, storage, and cloud functions allows the team to focus on core FoE logic such as pacts, verification flows, consequences. This way we can:

- Store user profiles, pacts, history of completed and failed pacts.
- Store media uploads, including photos and short videos, for both task completion and consequence execution.
- Run background checks at deadlines to determine whether to trigger consequences.
- Send push notifications for upcoming deadlines, pending verifications, and pact outcomes.

This supports real time behaviour and scaling for an MVP, at the cost of some provider lock in, which is acceptable at this stage.

AI and verification

FoE includes optional “AI referee” modes for tasks where users submit media evidence, and text features such as auto generated roast captions or summary checks. Instead of building complex AI from scratch, the team can use established APIs for text generation and basic classification or computer vision. These services help with:

- Generating witty but bounded roast captions and “tribute” posts for social consequences.
- Checking that written reflections or summaries meet a basic length and coherence threshold.
- Confirming simple properties of videos, such as duration, while leaving more nuanced judgment to the human friend.

Because user photos and videos involve sensitive data, advanced analysis such as detailed pose tracking can remain out of scope for the MVP. Verification in early versions should lean on friends as primary referees, with AI as a support layer rather than a strict gatekeeper.

Key technical challenges

- Handling frequent small media uploads with acceptable performance on mobile networks.
- Designing AI interactions that feel fair and are hard to exploit.
- Building clear permission and visibility controls so consequence posts stay within FoE’s environment and do not spill onto external platforms without consent.
- Clearly communicating data retention and visibility for any embarrassing or sensitive content.

Development plan for MVP

Estimated timeline for a 4-5 week part time student team:

Planning and UX design, Week 1

- Define MVP scope, including account creation, pact creation, friend invitation, basic consequence library, and a simple feed.
- Creating the prototype - figma.

Backend foundation, Week 1

- Set up authentication, user profiles, pact schema, pact history.
- Implement basic functions to handle deadlines and consequence resolution.

Mobile frontend core, Week 2

- Build onboarding, login, and profile screens.
- Implement pact creation including task, deadline, verification mode, consequence selection, and friend assignment.
- Build a home page

Verification and consequences, Week 3

- Support photo and video upload flows for proof and consequence evidence.
- Build friend review screens with accept and reject flows.
- Manage profile status changes based on pact outcomes.

Social and gamification layer, Week 3

- Implement a basic feed of pact outcomes and consequence posts.
- Add simple stats such as streaks, consistency score, and completion rate.
- Implement a basic in app consequence library unlocked or purchased with coins.

Testing and soft launch, Week 4

- Test with classmates and selected survey participants.
- Fix usability issues and refine consequence templates based on reactions and feedback.

Competitor Research

We reviewed three notable products related to habits, gamification, and commitment contracts to understand where FoE fits.

Habitica

Habitica treats tasks and habits as elements in a role playing game. Users gain experience and rewards by completing actions, and lose health by failing or missing tasks. Habitica has reached a significant user base and shows that gamification elements such as avatars, quests, and rewards support ongoing engagement when designed well (Trophy 2025, Habitica 2025). Users can form parties and cooperate on quests, which adds a social layer. The main stakes are virtual. When users lose interest in the game layer, they often lose motivation to keep tracking.

Beeminder

Beeminder is a commitment service where users define a goal and a “yellow brick road” of expected progress. If the user falls behind and “derails,” Beeminder charges a real money penalty to a linked payment method (Beeminder 2026, Productivity Directory 2024). This uses loss aversion very directly. It suits users who accept strong financial stakes and detailed data tracking. For many students, frequent financial penalties feel too stressful and expensive for day to day academic tasks.

StickK

StickK lets users create commitment contracts. Users set goals, stake money, and nominate a referee and supporters. If they fail, the staked money can go to a charity, an anti charity, or a friend (stickK 2025, Wikipedia 2008). This again leans on strong financial and reputational stakes. The platform is general purpose, aimed at a wide range of life goals, and feels formal and high stakes. For a student who wants help starting earlier on a report or study session, this level of commitment may feel excessive.

Across these tools, common patterns include community elements, streaks, points, and rewards for continued engagement (Vorecol 2024). Many products still treat accountability as either private tracking with light reminders, or as heavy financial enforcement. Few focus on playful, socially visible, but bounded consequences designed for student culture.

How FoE differentiates

FoE is positioned between game like habit trackers and money based commitment contracts.

- It adopts precommitment concepts from Beeminder and StickK, users agree in advance to specific consequences, but uses in app coins and safe social or effort consequences instead of large real money penalties.

- It uses gamification elements seen in Habitica, such as coins, badges, and streaks, but ties them to real social stakes through friend referees and a curated consequence library that affects status in the app.

- It focuses on students in high pressure academic environments, with consequences that are time bound and limited to the app, such as public apologies in feed, playful photos, physical challenges, or temporary profile changes. These matter to peers without creating lasting harm.

Existing tools show that gamification and commitment devices influence behaviour. There is still a gap for a student focused, socially enforced, non monetary accountability app that feels like a social platform rather than a financial contract or pure game. FoE aims to occupy this gap.

Market Research

We grounded FoE in current evidence on procrastination and student life in Singapore. Work on procrastination among university students in Singapore indicates high levels of procrastination and links it to stress, lower performance, and reduced wellbeing. Authors highlight the need for strategies that support earlier task initiation and more consistent habits (Yeo and Chiam 2025).

Singapore also has high smartphone use and internet access across young adults. Local commentary notes that students use mobile and social platforms heavily, including Instagram, TikTok, and Telegram, alongside academic demands (London School of Business and Finance 2025). A mobile first, social style app fits this media environment and behaviour. Instead of asking students to adopt slow or heavy workflows, FoE fits into existing patterns, such as checking feeds, posting updates, and reacting to friends.

The academic culture in Singapore emphasises performance, grades, and competition. At the same time, students often juggle part time work, co curricular activities, and social commitments (London School of Business and Finance 2025). This increases both the costs of procrastination and the value of tools that help protect focused time. FoE's near term consequences and visible follow through support students who want more structure but struggle to maintain it alone.

Target market, initial

Primary focus:

- University undergraduates in Singapore aged roughly 17–25 at institutions such as NUS, NTU, SMU, SIM, and the polytechnics.

- Tech savvy, active on social platforms, frequent procrastinators, often engage in last minute cramming.

- Motivated by peer perception and social validation, responsive to gamified structures.

In the longer term, FoE can expand into similar high pressure academic markets in East and Southeast Asia, for example Hong Kong, South Korea, and Japan, and into international student segments that share similar pressures (London School of Business and Finance 2025).

Business and product direction

FoE will use a freemium model. Core accountability features stay free so students with limited budgets still gain value. A premium tier around S$2.99 to S$4.99 per month would add unlimited pacts, richer analytics, exclusive consequences, and cosmetic upgrades. Partnerships with brands popular among students, such as food delivery or cafes, could support sponsored rewards and make the in app coin system feel more concrete (future implementation, second iteration).

Early growth will likely come from campus networks and friend referrals because each pact needs at least one “Foe” (if choose friend). Over time, FoE can explore collaboration with university offices focused on student life or wellbeing as an optional productivity aid, while avoiding positioning as a clinical mental health tool.

The long term goal is for FoE to act as an accountability layer that sits on top of calendars and to do lists. Students still use their existing tools to plan tasks, but use FoE when they need extra pressure and social commitment to follow through.

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